

JDE Volume and Price Conversion Logic

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A complete, self-contained guide to converting product volume into short tons across JDE Cost Analyzer, Sales Order, and General Ledger data. Written so a new analyst can follow it with no prior JDE knowledge.

1. Background and Business Rationale

1.1 What we are doing

We convert product volume into short tons (TN) as the single, common reporting measure across three JD Edwards data sources. Each source records volume in a different business context: Cost Analyzer (F1602) for managerial and budget versus actual reporting, Sales Order Detail (F4211) for open orders and shipments, and Account Ledger (F0911) for posted general ledger activity. All three use one shared conversion method so a ton means the same thing everywhere.

1.2 Why it matters

- **Products sell in mixed units.** Sand ships in pounds, bags, super sacks, and tons. You cannot add or compare pounds plus bags plus tons. Tons is the physical common denominator for an industrial minerals business, and it is the unit both Oil and Gas and Industrial and Specialty Products report on.
- **The business runs on per-ton metrics.** Tons shipped, price per ton, revenue per ton, cost per ton, lane profitability, plant throughput. Sales and Finance decide in dollars per ton, so volume must land in tons before that math works.
- **Three systems must reconcile.** Delivered orders (F4211), actuals and budget (F1602), and posted revenue (F0911) have to tie to the same tons. If each source converted its own way, a ton on a dashboard would not match a ton in the ledger, and the numbers lose trust.

1.3 Why the logic is built the way it is

- **Item-specific before standard.** F41002 carries the real packaging weight for a given item, which differs by product. F41003 is the generic fallback. Item-specific must win, or packaging weight gets averaged away.
- **Scaling divisions.** JDE stores quantities and factors as whole numbers with implied decimals. Un-scaling returns real values before multiplying. See Section 2.3.
- **Sign handling.** Cost Analyzer separates budget and actual and stores direction by ledger. The convention in the current F1602 build is BU and BA positive, AU and AA negative, which flips actual rows to read positive. See Section 9.4.
- **Bidirectional lookup with inversion.** Some items are set up only as TN to unit. The lookup must resolve the unit to TN factor in either orientation, and when it resolves from a reverse row it must invert the stored factor. See Section 7.
- **Missing-conversion flag.** If an item has no factor, tons would silently default to raw quantity and corrupt totals. The flag turns a silent data error into a visible Item Master setup task.

1.4 The outcome

One trustworthy tons figure that flows from open orders, through shipments, to posted GL revenue. That is what lets dollars-per-ton pricing analytics, volume forecasting, lane profitability, and budget versus actual all reconcile to the same number.

2. JDE Concepts You Need First

Read this once if you are new to JDE. Everything later in the document builds on these five ideas.

2.1 Tables are named with an F prefix

JD Edwards EnterpriseOne is the ERP system. Its data lives in tables named with a leading F, short for file. F4211 is the Sales Order Detail table, F1602 is Cost Analyzer Balances, F0911 is the Account Ledger.

2.2 Every column starts with the table's two-letter prefix

Each table's columns share a short prefix, so you can tell at a glance which table a field belongs to.

Prefix	Table	Example column
SD	F4211 Sales Order Detail	SDITM = item, SDUOM = unit of measure
CB	F1602 Cost Analyzer Balances	CBITM = item, CBLT = ledger type
GL	F0911 Account Ledger	GLITM = item, GLU = units, GLUM = unit of measure
IM	F4101 Item Master	IMITM = item, IMUOM1 = primary unit
UM	F41002 Item UoM Conversion	UMCONV = conversion factor
UC	F41003 Standard UoM Conversion	UCCONV = conversion factor

2.3 Numbers are stored as scaled whole numbers

To avoid rounding errors, JDE stores numbers without a decimal point and defines the number of decimals separately. To get a real value you divide by the scale. This is why the conversion formulas divide.

Value	Where	Divide by	Real value
Sales quantity	F4211 SDUORG, SDSOQS, SDPQOR	1,000	stored / 1,000
Cost / GL quantity	F1602 CBNBxx, F0911 GLU	100	stored / 100
Unit price	F4211 SDUPRC	1,000,000	stored / 1,000,000
Conversion factor	F41002 UMCONV, F41003 UCCONV	10,000,000	stored / 10,000,000

Rule of thumb: if a JDE number looks far too large, it is scaled. Divide by its scale before using it. The conversion factor scale of 10,000,000 (7 decimals) is standard. The precision, the number of implied decimals for each column, is provided by the etlconfig / meta table per column per table, so the load reads the scale from metadata rather than hardcoding it. The divide-by values shown here are the current known values for this environment.

2.4 Items carry a Unit of Measure

A Unit of Measure (UoM) is how an item is counted or weighed, for example LB (pounds), BG (bag), or TN (short ton). Every item has a primary UoM in Item Master (IMUOM1). Sales order lines carry their own transaction UoM (SDUOM) and pricing UoM (SDUOM4). GL lines carry their own unit (GLUM). Cost Analyzer has no line UoM and uses the item primary UoM.

The UoM the conversion lookup matches on is called the Match UoM: IMUOM1 for F1602, SDUOM (volume) and SDUOM4 (price) for F4211, and GLUM for F0911.

2.5 Ledger Type separates volume from dollars

In Cost Analyzer and the GL, each row is tagged with a Ledger Type telling you whether it holds volume or money. Only volume rows get converted to tons. Dollar rows are left alone.

Ledger	Meaning	Holds
AU	Actual Units	Volume, converted to tons
BU	Budget Units	Volume, converted to tons
AA	Actual Amount	Dollars, not converted
BA	Budget Amount	Dollars, not converted

Subledger note (F1602 and F0911): volume conversion excludes rows where Subledger Type = 'X' (CBSBLT for F1602, GLSBLT for F0911). 'X' rows are not valid item-level volume and would overstate tons. The exclusion applies to the volume path only, dollar totals are unaffected. F4211 is not affected.

Item link note (F0911): the item is read directly from GLITM and the UoM match uses the row's own unit GLUM. No Subledger Type filter is used to reach the item. GLSBL and GLSBLT are passthrough attributes only.

3. Purpose and Scope

Purpose. Define one standard, repeatable method to convert product volume into short tons so every report and data model uses a single common scale, regardless of the JDE source table.

Scope. Any report, view, or model that reads a quantity from F1602, F4211, or F0911 and needs the result in tons, plus price per ton for F4211. Dollar rows are out of scope, they are never multiplied by a tons factor.

4. How to Read This Document

- **Brand new:** read Sections 1 and 2 first, they need no prior knowledge.
- **Business users:** Sections 1, 5, 7, 8, and 11.
- **Techno-functional:** Sections 6, 9, 10, and 14 for build and validation.

Throughout, **[HUBBLE TODAY]** marks current behavior and its limitations, and **[POWER BI TARGET]** marks the standard we are building in the migration.

Report authors: the required fields to add to every volume or price report are listed in Section 9.1, with the per-table grain and expressions in Section 10.4.

5. Key Terms

- **Selling unit:** the unit an item is transacted in, for example LB, BG, or TN.
- **Conversion factor:** tons per one selling unit, set up once per item, looked up automatically.
- **Volume row:** a quantity row. A dollar row is never converted.
- **Short tons (TN):** the common reporting scale all volumes are converted to, 2000 lb.
- **Cascade:** the fixed order in which the factor is looked up, item-specific first, then standard, then default.

6. Table and Field Glossary

6.1 Conversion and Reference Tables

JDE Name	Alias	Description
F41002	Item UoM Conversion	Item-specific unit of measure conversion factors. Tried first.
UMITM	Item Number - Short	Numeric item id, join key.

JDE Name	Alias	Description
UMUM	From Unit of Measure	Source unit of the conversion row.
UMRUM	To Unit of Measure	Target unit, filtered to TN.
UMCONV	Conversion Factor (raw)	7 implied decimals. Real factor = UMCONV / 10,000,000.
F41003	Standard UoM Conversion	Item-independent standard conversion. Fallback when no item row exists.
UCUM	From Unit of Measure	Source unit.
UCRUM	To Unit of Measure	Target unit, filtered to TN.
UCCONV	Conversion Factor (raw)	7 implied decimals. Real factor = UCCONV / 10,000,000.
F4101	Item Master	Supplies each item's primary selling unit.
IMITM	Item Number - Short	Numeric item id.
IMLITM	Second Item Number	Alternate item number.
IMUOM1	Unit of Measure - Primary	Item primary UoM, used for F1602 matches.
IMDSC1	Item Description	Item name.

6.2 Source Transaction Tables

JDE Name	Alias	Description
F1602	Cost Analyzer Balances	Monthly volume and amount by item, plant, ledger.
CBITM	Item Number - Short	Item link to F4101.
CBLT	Ledger Type	AU, BU volume; AA, BA dollars.
CBSBLT	Subledger Type	'X' rows excluded from volume conversion.
CBNB01..CBNB12	Period 1..12 Net Balance	Monthly buckets. Unpivot to one value per month.
CBABR4	Branch Plant Number	Plant reference.
F4211	Sales Order Detail	Order line level shipped and open quantities.
SDITM	Item Number - Short	Item link, direct to F41002.
SDUOM	Transaction UoM	Line transaction unit, used for the tons match.
SDUOM4	Pricing UoM	Pricing unit, used for the price-per-ton match.
SDUPRC	Unit Price (raw)	Stored / 1,000,000.
SDUORG / SDSOQS / SDPQOR	Transaction / Shipped / Priced Qty	Volume quantities. Stored / 1,000.
SDLNID	Line Number	Line identity. Stored / 1,000.
SDDOC / SDDCT	Invoice Number / Type	Populated when invoiced. Basis for the delivered scope, with status codes.
SDMCU	Branch Plant Number	Plant reference.
F0911	Account Ledger	GL line with units and amounts.
GLITM	Item Number - Short	Item link, direct to F41002.
GLUM	Unit of Measure	Row unit, used for the tons match.
GLU	Units	Volume quantity. Stored / 100.
GLAA	Amount	Dollar amount. Not converted.
GLLT	Ledger Type	AU units, AA amount.
GLSBL / GLSBLT	Subledger / Type	Passthrough. 'X' rows excluded from volume conversion.
GLMCU	Business Unit	Cost center.

7. How the Conversion Works

Every item sells in its own unit. To report everything on one scale, multiply the item's volume by a tons-per-unit factor. The factor is looked up automatically, first match wins.

Resolution order for target TN, item I, lookup unit U:

- **Already tons:** if U = 'TN', factor = 1.
- **F41002 direct:** row UMUM = U, UMRUM = 'TN' on item I, factor = UMCONV / 10,000,000.
- **F41002 inverse:** else row UMUM = 'TN', UMRUM = U on item I, factor = 10,000,000 / UMCONV.
- **F41003 direct:** else row UCUM = U, UCRUM = 'TN', factor = UCCONV / 10,000,000.
- **F41003 inverse:** else row UCUM = 'TN', UCRUM = U, factor = 10,000,000 / UCCONV.
- **Default:** else factor = 1 and raise the missing-conversion flag.

The resolved factor is always tons per one unit, so volume multiplies and price divides regardless of how the row was set up.

The cascade, in SQL:

```
[Conversion Factor] =  
COALESCE(  
  CASE WHEN [Match UoM] = 'TN' THEN 1 END,    -- already tons  
  cn1_direct.[Factor],    -- F41002 UMCONV/1e7  
  cn1_inverse.[Factor],   -- F41002 1e7/UMCONV  
  cn2_direct.[Factor],    -- F41003 UCCONV/1e7  
  cn2_inverse.[Factor],   -- F41003 1e7/UCCONV  
  1)                      -- default, flag as missing  
  
-- [Match UoM] is the UoM the lookup matches on, by source:  
-- F1602 -> IMUOM1 (item primary, F1602 has no line UoM)  
-- F4211 -> SDUOM (volume), SDUOM4 (price)  
-- F0911 -> GLUM (row unit)
```

The tons formula:

```
Volume (Tons) = ( Raw Quantity / Quantity Scale )  
               * Sign * [Conversion Factor]
```

Missing-conversion flag: set to Y when the cascade returns nothing before the default 1. Any row flagged Y has unreliable tons and points to an Item Master setup task.

Hubble limitation: the current Hubble queries for F1602, F4211, and F0911 use item-specific F41002 direct only. There is no F41003 fallback, no inverse handling, and no missing-conversion flag, so an item without an F41002 row silently defaults to factor 1. Hubble also carries UMCONV as a passthrough column and applies the multiply downstream. Closing these gaps in-model is a primary driver of the Power BI migration.

8. Pricing UoM Conversion (F4211)

Sales order lines carry a separate pricing unit of measure (SDUOM4) and a unit price (SDUPRC). To report price per ton, the per-unit price is converted using the same F41002 then F41003 cascade, matched on the pricing UoM. This applies to F4211 only, since only sales lines hold a price. **[POWER BI TARGET]**

Scope, delivered orders only. Pricing UoM conversion and price per ton apply to delivered lines only. Delivered means invoiced. Open orders use separate logic that is out of scope of this document. Delivered filter: Last Status <> 980 and Next Status = 999.

8.1 The key difference: divide, do not multiply

The conversion factor is tons per one selling unit. Volume is a quantity in selling units, so multiplying gives tons. Price is dollars per selling unit, so it must be divided by the factor to become dollars per ton.

Same factor, opposite operation. Volume: $\text{quantity} \times \text{factor} = \text{tons}$. Price: $\text{price} / \text{factor} = \text{price per ton}$. Dollars per unit, divided by tons per unit, equals dollars per ton.

8.2 Pricing cascade and formula

The lookup order mirrors the volume cascade, matched on SDUOM4 instead of SDUOM, with the same inverse and F41003 handling.

```
Price per Ton = ( SDUPRC / 1,000,000 )  
                / NULLIF([Pricing Conversion Factor], 0)  
-- delivered lines only: Last Status <> 980 AND Next Status = 999
```

Missing price-conversion flag: set to Y when the pricing cascade returns nothing before the default 1. A Y means price per ton for that row is unreliable.

Hubble limitation: Hubble does not produce price per ton. The F4211 query carries SDUOM4 and SDUPRC as raw passthrough columns with no pricing cascade and no per-ton price. Delivering price per ton is a core objective of the migration.

9. Source Differences

The cascade is identical for all three sources. What changes is the UoM used for the match, the item link, the quantity scale, the sign, and the volume filter.

Parameter	F1602 (Cost Analyzer)	F4211 (Sales Order)	F0911 (Account Ledger)
Conversion cascade	F41002 then F41003, inverse, flag	F41002 then F41003, inverse, flag	F41002 then F41003, inverse, flag
UoM used for match	Item primary (IMUOM1)	Line transaction (SDUOM), pricing (SDUOM4)	Row unit (GLUM)
Item link	CBITM	SDITM (direct)	GLITM (direct)
Quantity field	CBNB01..CBNB12	SDUORG, SDSOQS, SDPQOR	GLU
Quantity scale	/ 100	/ 1,000	/ 100
Sign	BU/BA +1, AU/AA -1	Inherent in document	None, native ledger sign, no flip
Volume filter	Subledger Type <> 'X'	None	Subledger Type <> 'X'
Price per ton	N/A, no unit price	Yes, on SDUOM4	N/A, no unit price

Key nuance: F4211 and F0911 both carry their own line UoM and match F41002 on it (SDUOM, GLUM). Only F1602 has no line UoM and matches on the item primary UoM (IMUOM1).

9.1 Required Output Columns per Table

Every model built on these sources must expose the columns below. Price per ton is F4211 only. **[POWER BI TARGET]**

F1602 Cost Analyzer Balances

Required field	Source and derivation
Source Volume	CBNB01..CBNB12 unpivot, de-scaled value before conversion.
Source UoM (Volume)	IMUOM1 from F4101 (join CBITM to IMITM). F1602 has no line UoM.
Conversion Factor (Volume)	Cascade on IMUOM1 to TN, F41002 then F41003, inverse, flag.

Required field	Source and derivation
Converted Volume (Tons)	(CBNBxx / scale) * Sign * Factor. BU/BA +1, AU/AA -1. AU and BU only, exclude CBSBLT = 'X'.
Price fields	N/A. No per-unit price stored on F1602.

F4211 Sales Order Detail

Required field	Source and derivation
Source Volume	SDUORG or SDSOQS or SDPQOR, de-scaled value before conversion.
Source UoM (Volume)	SDUOM, transaction UoM at the line.
Conversion Factor (Volume)	Cascade on SDUOM to TN, item link SDITM direct.
Converted Volume (Tons)	(SDUORG / scale) * Factor.
Source Price	SDUPRC, de-scaled value before conversion.
Source UoM (Price)	SDUOM4, pricing UoM at the line.
Conversion Factor (Price)	Cascade on SDUOM4 to TN.
Converted Price per Ton	(SDUPRC / scale) / NULLIF(Price Factor, 0). Delivered lines only, Last Status <> 980 and Next Status = 999.

F0911 Account Ledger

Required field	Source and derivation
Source Volume	GLU, de-scaled value before conversion.
Source UoM (Volume)	GLUM, the row's own unit. F0911 matches F41002 on GLUM.
Conversion Factor (Volume)	Cascade on GLUM to TN, item link GLITM direct.
Converted Volume (Tons)	(GLU / scale) * Factor. Units ledger (AU) only, exclude GLSBLT = 'X'. No sign flip, native ledger sign preserved.
Price fields	N/A. Amounts (GLAA) and units (GLU) post separately, no per-unit price.

Note on dollars per ton for F1602 and F0911. These sources have no stored per-unit price to convert. If a per-ton dollar figure is needed, derive it downstream as dollar-ledger value divided by tons, which is a computed unit economic, not a price-UoM conversion.

9.2 Hubble Limitations vs Power BI Target

Capability	Hubble today	Power BI target
Item-specific F41002	Yes, direct only	Yes
F41003 standard fallback	No	Yes, required
Inverse orientation	No	Yes, required
Missing-conversion flag	No, silent default 1	Yes
Volume to tons in model	No, applied downstream	Yes
Price per ton (F4211)	No	Yes, required
Subledger 'X' exclusion	Applied in report layer	In model, F1602 and F0911

9.3 Volume Filter

For F1602 and F0911, volume conversion excludes rows where Subledger Type = 'X'. The exclusion applies to the unit ledgers only and does not affect dollar totals. F4211 has no such filter.

9.4 Sign Rule

Signs follow the current F1602 build: BU and BA multiply by +1, AU and AA multiply by -1, which flips actual rows to read positive. F4211 sign is inherent in the document. F0911 does not need a sign change like F1602, it preserves the native ledger sign with no flip. Confirm signs against a known item during validation.

10. Join Details by Source

Each block states the table grain and returns the six standard columns: Volume UoM, Price UoM, Volume Factor, Price Factor, Volume Tons, Price per Ton. The direct and inverse F41002 and F41003 joins implement the Section 7 resolver. 1e7 = 10,000,000.

10.1 F1602 Cost Analyzer Balances

```
-- Grain (primary key): CBVWNM, CBACTB, CBAID, CBCFY, CBLT, CBSBL,
-- CBSBLT, CBABR1, CBABT1 .. CBABR4, CBABT4, CBITM
-- Then unpivot CBNB01..CBNB12 -> one row per fiscal month
SELECT
  i.IMUOM1 AS [Volume UoM], -- item primary, no line UoM
  CAST(NULL AS NVARCHAR(4)) AS [Price UoM], -- N/A, no price on F1602
  COALESCE(CASE WHEN i.IMUOM1='TN' THEN 1 END,
    v1d.UMCONV/1e7, 1e7/NULLIF(v1i.UMCONV,0),
    v2d.UCCONV/1e7, 1e7/NULLIF(v2i.UCCONV,0), 1) AS [Volume Factor],
  CAST(NULL AS FLOAT) AS [Price Factor], -- N/A
  (m.[Value]/100)
    * CASE WHEN t.CBLT IN ('BU','BA') THEN 1 ELSE -1 END
    * COALESCE(CASE WHEN i.IMUOM1='TN' THEN 1 END,
      v1d.UMCONV/1e7, 1e7/NULLIF(v1i.UMCONV,0),
      v2d.UCCONV/1e7, 1e7/NULLIF(v2i.UCCONV,0), 1) AS [Volume Tons],
  CAST(NULL AS FLOAT) AS [Price per Ton] -- N/A
FROM PRODDTA.F1602 t
LEFT JOIN PRODDTA.F4101 i ON t.CBITM = i.IMITM
LEFT JOIN PRODDTA.F41002 v1d
  ON v1d.UMITM=i.IMITM AND v1d.UMUM=i.IMUOM1 AND v1d.UMRUM='TN'
LEFT JOIN PRODDTA.F41002 v1i
  ON v1i.UMITM=i.IMITM AND v1i.UMUM='TN' AND v1i.UMRUM=i.IMUOM1
LEFT JOIN PRODDTA.F41003 v2d ON v2d.UCUM=i.IMUOM1 AND v2d.UCRUM='TN'
LEFT JOIN PRODDTA.F41003 v2i ON v2i.UCUM='TN' AND v2i.UCRUM=i.IMUOM1
/* m = CBNB01..CBNB12 unpivoted to [Value] */
WHERE t.CBLT IN ('AU','BU') AND t.CBSBLT <> 'X'
```

10.2 F4211 Sales Order Detail

```
-- Grain (primary key): SDKCOO, SDDOCO, SDDCTO, SDLNID
-- (order company, order number, order type, line number)
SELECT
  t.SDUOM AS [Volume UoM],
  t.SDUOM4 AS [Price UoM],
  COALESCE(CASE WHEN t.SDUOM='TN' THEN 1 END,
    vd.UMCONV/1e7, 1e7/NULLIF(vi.UMCONV,0),
    sd.UCCONV/1e7, 1e7/NULLIF(si.UCCONV,0), 1) AS [Volume Factor],
  COALESCE(CASE WHEN t.SDUOM4='TN' THEN 1 END,
    pd.UMCONV/1e7, 1e7/NULLIF(pi.UMCONV,0),
    qd.UCCONV/1e7, 1e7/NULLIF(qi.UCCONV,0), 1) AS [Price Factor],
  (t.SDSOQS/1000)
    * COALESCE(CASE WHEN t.SDUOM='TN' THEN 1 END,
      vd.UMCONV/1e7, 1e7/NULLIF(vi.UMCONV,0),
      sd.UCCONV/1e7, 1e7/NULLIF(si.UCCONV,0), 1) AS [Volume Tons],
  (t.SDUPRC/1000000)
    / NULLIF(COALESCE(CASE WHEN t.SDUOM4='TN' THEN 1 END,
      pd.UMCONV/1e7, 1e7/NULLIF(pi.UMCONV,0),
      qd.UCCONV/1e7, 1e7/NULLIF(qi.UCCONV,0), 1),0) AS [Price per Ton]
```



```

FROM PRODDTA.F4211 t
-- volume factor on SDUOM
LEFT JOIN PRODDTA.F41002 vd
  ON vd.UMITM=t.SDITM AND vd.UMUM=TRIM(t.SDUOM) AND vd.UMRUM='TN'
LEFT JOIN PRODDTA.F41002 vi
  ON vi.UMITM=t.SDITM AND vi.UMUM='TN' AND vi.UMRUM=TRIM(t.SDUOM)
LEFT JOIN PRODDTA.F41003 sd ON sd.UCUM=TRIM(t.SDUOM) AND sd.UCRUM='TN'
LEFT JOIN PRODDTA.F41003 si ON si.UCUM='TN' AND si.UCRUM=TRIM(t.SDUOM)
-- price factor on SDUOM4
LEFT JOIN PRODDTA.F41002 pd
  ON pd.UMITM=t.SDITM AND pd.UMUM=TRIM(t.SDUOM4) AND pd.UMRUM='TN'
LEFT JOIN PRODDTA.F41002 pi
  ON pi.UMITM=t.SDITM AND pi.UMUM='TN' AND pi.UMRUM=TRIM(t.SDUOM4)
LEFT JOIN PRODDTA.F41003 qd ON qd.UCUM=TRIM(t.SDUOM4) AND qd.UCRUM='TN'
LEFT JOIN PRODDTA.F41003 qi ON qi.UCUM='TN' AND qi.UCRUM=TRIM(t.SDUOM4)

```

10.3 F0911 Account Ledger

```

-- Grain (primary key): GLKCO, GLDCT, GLDOC, GLDGJ, GLJELN, GLLT, GLEXTL
SELECT
  a.GLUM AS [Volume UoM], -- row unit
  CAST(NULL AS NVARCHAR(4)) AS [Price UoM], -- N/A, no price on F0911
  COALESCE(CASE WHEN a.GLUM='TN' THEN 1 END,
    vd.UMCONV/1e7, 1e7/NULLIF(vi.UMCONV,0),
    sd.UCCONV/1e7, 1e7/NULLIF(si.UCCONV,0), 1) AS [Volume Factor],
  CAST(NULL AS FLOAT) AS [Price Factor], -- N/A
  (a.GLU/100) -- no sign flip on F0911
  * COALESCE(CASE WHEN a.GLUM='TN' THEN 1 END,
    vd.UMCONV/1e7, 1e7/NULLIF(vi.UMCONV,0),
    sd.UCCONV/1e7, 1e7/NULLIF(si.UCCONV,0), 1) AS [Volume Tons],
  CAST(NULL AS FLOAT) AS [Price per Ton] -- N/A
FROM PRODDTA.F0911 a
LEFT JOIN PRODDTA.F41002 vd
  ON vd.UMITM=a.GLITM AND vd.UMUM=a.GLUM AND vd.UMRUM='TN'
LEFT JOIN PRODDTA.F41002 vi
  ON vi.UMITM=a.GLITM AND vi.UMUM='TN' AND vi.UMRUM=a.GLUM
LEFT JOIN PRODDTA.F41003 sd ON sd.UCUM=a.GLUM AND sd.UCRUM='TN'
LEFT JOIN PRODDTA.F41003 si ON si.UCUM='TN' AND si.UCRUM=a.GLUM
WHERE a.GLLT = 'AU' AND a.GLSBLT <> 'X' -- units path, exclude 'X'
-- Item GLITM direct, match on GLUM, native ledger sign (no flip)
-- scale per column from etlconfig / meta table

```

Bidirectional setup: the conversion resolves the unit to TN factor in either orientation. When it resolves from a reverse row (TN to unit), it inverts the stored factor, 10,000,000 / UMCONV or 10,000,000 / UCCONV.

10.4 Grain and Output Expressions per Table

Element	F1602	F4211	F0911
Grain (primary key)	CBVWNM, CBACTB, CBAID, CBCFY, CBLT, CBSBL, CBSBLT, CBABR1-4, CBABT1-4, CBITM	SDKCOO, SDDOCO, SDDCTO, SDLNID	GLKCO, GLDCT, GLDOC, GLDGJ, GLJELN, GLLT, GLEXTL
Source UoM (Volume)	IMUOM1	SDUOM	GLUM
Source UoM (Price)	N/A	SDUOM4	N/A
Conversion Factor (Volume)	cascade IMUOM1 to TN	cascade SDUOM to TN	cascade GLUM to TN
Conversion Factor (Price)	N/A	cascade SDUOM4 to TN	N/A
Volume (Tons)	(CBNBxx / scale) * Sign * Factor	(SDUORG / scale) * Factor	(GLU / scale) * Factor, no sign flip

Element	F1602	F4211	F0911
Price per Ton	N/A	(SDUPRC / scale) / Price Factor, delivered only	N/A

Precision: the divide scale for each column is provided by the etlconfig / meta table per column per table, not hardcoded. The numeric scales shown are the current known values for this environment.

11. Worked Examples

All examples use short tons, 2000 lb.

11.1 F1602, Actual Units ledger (AU)

- Item sold in LB. Short ton factor = $1 / 2000 = 0.0005$ tons per lb.
- CBNBxx raw -441,700,000, un-scaled = $-441,700,000 / 100 = -4,417,000$
- Ledger AU, Sign = -1
- Tons = $-4,417,000 \times -1 \times 0.0005 = 2,208.50$**

Data caution: some LB rows in F41003 encode a factor near 0.00045359, which is a metric tonne (2204.62 lb), not a short ton. On the short ton basis the LB factor must be 0.0005. Verify LB to TN rows against the intended ton before load.

11.2 F4211, bagged item shipped (volume)

- Item sold in 50 lb bags (BG). Factor = $50 / 2000 = 0.025$ tons per bag.
- SDSOQS raw 40,000,000, un-scaled = $40,000,000 / 1,000 = 40,000$ bags
- Sign inherent, positive shipment
- Tons = $40,000 \times 0.025 = 1,000.00$**

11.3 F4211, price per ton (pricing UoM)

- Same item priced per bag. Pricing UoM SDUOM4 = BG, factor = 0.025 tons per bag
- SDUPRC raw 2,500,000, un-scaled = $2,500,000 / 1,000,000 = 2.50$ dollars per bag
- Price per Ton = $2.50 / 0.025 = 100.00$** (divide, not multiply)

11.4 F0911, units posted in pounds

- Row unit GLUM = LB. Short ton factor = 0.0005.
- GLU raw 200,000,000, un-scaled = $200,000,000 / 100 = 2,000,000$ (scale from etlconfig)
- No sign change, native ledger sign preserved
- Tons = $2,000,000 \times 0.0005 = 1,000.00$**

12. Validation

- Row orientation:** confirm reverse-direction rows apply $1 / \text{factor}$. A unit-to-TN result that looks like a units-per-ton figure indicates a missing inversion.
- Fallback order:** F41003 fires only when no F41002 item row exists. Confirm no double counting.
- Uniqueness:** F41002 must return one row per item plus From UoM after the TN filter. Duplicates inflate tons.

- **Missing-conversion flag:** a Y means the default 1 was applied. Tons for that row are unreliable, route the item for setup.
- **Price flag:** a Y on the price-conversion flag means price per ton could not be derived. Treat it as unreliable.
- **Volume only:** if a dollar total shifts after conversion, an amount row was converted in error.
- **Subledger 'X':** confirm 'X' rows are excluded from tons for F1602 and F0911, and that dollar totals are unchanged by the filter.
- **Sign:** confirm signs against a known item. F1602 BU/BA +1, AU/AA -1. F0911 uses the native ledger sign with no flip.
- **Precision:** confirm each column's divide scale is read from the etlconfig / meta table, not hardcoded.
- **Ton basis:** confirm LB to TN rows encode a short ton, 0.0005, not a metric tonne, 0.00045359.
- **Divide vs multiply:** volume multiplies by the factor, price divides by it. Swapping the two is the most common error.
- **Spot check:** tie one known item to a source-system inquiry.

13. Exceptions and Edge Cases

- **No conversion row:** an item not in TN with no F41002 and no F41003 row cannot be converted. Flag for Item Master setup.
- **Mixed UoM (F4211):** one item with different UoM across lines is converted per line using the line UoM, never a single blended factor.
- **Non-item GL rows (F0911):** rows with no resolvable GLITM carry no item and cannot be converted.
- **Subledger 'X' rows:** excluded from volume conversion for F1602 and F0911.
- **Open orders (F4211):** lines with no invoice have volume in tons but no price per ton by design. Delivered filter: Last Status <> 980 and Next Status = 999.

14. Quick Reference

14.1 Scaling cheat sheet

Value	Column	Divide by	Applies to
Sales quantity	SDUORG, SDSOQS, SDPQOR	1,000	F4211
Cost / GL quantity	CBNB01..CBNB12, GLU	100	F1602, F0911
Unit price	SDUPRC	1,000,000	F4211
Conversion factor	UMCONV, UCCONV	10,000,000	F41002, F41003
Line number	SDLNID	1,000	F4211

Precision source: these divide-by values come from the etlconfig / meta table, defined per column per table. The values shown are the current known scales for this environment.

14.2 Volume vs price direction

Measure	Operation	Result
Volume (quantity)	Quantity x Conversion Factor	Tons
Price (per unit)	Price / Pricing Conversion Factor	Price per ton

14.3 Ledger types

Code	Meaning	Convert to tons
AU	Actual Units	Yes
BU	Budget Units	Yes
AA	Actual Amount (dollars)	No
BA	Budget Amount (dollars)	No

14.4 Sign rule by source

Source	Sign
F1602 Cost Analyzer	BU and BA = +1, AU and AA = -1
F4211 Sales Order	Inherent in the document, no override applied
F0911 Account Ledger	None, native ledger sign preserved, no flip

14.5 Common units of measure

UoM codes are defined in JDE user defined codes (UDC 00/UM). Common examples below, confirm against Item Master for a given item.

Code	Unit
TN	Short ton, 2000 lb (the reporting target)
LB	Pound
BG	Bag
CWT	Hundredweight (100 lb)
EA	Each